NEST: Locally Certified Search for Structural-Entropy Hierarchies

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Abstract. Structural entropy evaluates a graph hierarchy through the description length of an encoding tree. HCSE and BBM [10] construct such trees but do not determine whether the result admits a local entropy-reducing topology change. We introduce NNI-certified Entropy Search over Trees (Nest), a multi-start optimizer based on rooted nearest-neighbor interchange (NNI). For weighted graphs, we derive the exact NNI entropy change and prove that an improving rotation can be inaccessible to a monotone merge or compression. The formula supports best-improvement descent and a checkable one-NNI local-optimality certificate. Because a better tree may require a temporary increase, Nest also tests bounded two-move paths but commits only improving endpoints. Using known binary sufficiency [12, 10], we formulate an $O(3^n)$ arbitrary-subset specialization of trellis inference and audit global optima on small graphs. Across 500 graphs from five 64-vertex hierarchical regimes, Nest lowers entropy by 0.4239 ± 0.0086 bits relative to the better of HCSE and BBM and wins all 500 paired comparisons. In sealed, candidate-budget-matched audits, 32 coalescent starts reach the exact global optimum on 249/250, 99/100, and 22/25 instances at $n = 12, 14, 16$, respectively (370/375 overall), whereas 32-call HCSE and label-free BBM reach 2/375 and 1/375. These exact-instance results certify observed optimality on the declared suites without asserting a general approximation ratio.

Keywords: Structural entropy · hierarchical clustering · tree editing · nearest-neighbor interchange · graph algorithms

1 Introduction

Many networks are organized at more than one scale: vertices form local groups, related groups form larger modules, and the desired output is therefore a tree rather than one flat partition. Structural entropy supplies an information-theoretic objective for such an encoding tree [6]. Recent HCSE and BBM methods construct low-entropy hierarchies through recursive partitioning and balanced merging [10]. Their construction rules, however, do not answer a complementary question: *after a tree has been constructed, does a small topology change still lower the same objective?*

Construction and local optimality are distinct: early decisions can retain avoidable entropy. We therefore seek a topology operator whose effect is exact, whose termination is interpretable, and whose output can be compared with a true optimum on instances small enough for exhaustive certification.

Nearest-neighbor interchange (NNI) is the elementary rotation of a rooted binary tree. It changes one internal edge while preserving every leaf. NNI is standard in phylogenetic tree search and has been studied for other hierarchical-clustering objectives, but its structural-entropy change has not been used as an exact graph-hierarchy optimizer. We show that, for $((A, B), C) \to (A, (B, C))$, all objective terms cancel except two cross-module edge weights and three module volumes. This identity yields monotone descent and a one-NNI local-optimality certificate.

Our method, NNI-certified Entropy Search over Trees (Nest), refines several inexpensive initial trees with the exact operator and returns the lowest independently verified endpoint. A bounded two-move beam can cross a small intermediate barrier, but accepts a pair only when its endpoint lowers the full objective. A randomized-restart variant, Nest-R, increases basin coverage without inspecting labels or the exact optimum. Using prior binary sufficiency [12, 10], we specialize a subset-trellis dynamic program to compute the global optimum on small graphs; binary sufficiency itself is not our contribution. It separates three claims that are often conflated: monotone improvement, local certification, and global optimality.

Our contributions are:

- an exact weighted-graph NNI delta, a proof of separation from monotone single merge/compress steps, and a checkable one-NNI local certificate;
- Nest, which combines multi-start construction, exact NNI refinement, and bounded two-move escape without weakening the output guarantee;
- an SE-specific $O(3^n)$ arbitrary-subset trellis optimizer for small graphs, built on prior binary sufficiency, and an exact audit of Nest's optimality gap; and
- a paired evaluation against SE agglomeration, Louvain-2L, Paris, and the official HCSE and BBM implementations, with entropy, hierarchy recovery, runtime, memory, operator audits, and component ablations.

Across five frozen regimes, Nest lowers entropy by 0.5468 ± 0.0218 bits versus HCSE and 0.5213 ± 0.0101 bits versus BBM, winning every paired comparison across 500 graphs. Across sealed exact suites at $n = 12, 14, 16$, 32 coalescent Nest starts reach the global optimum in 370/375 cases, compared with 2/375 for 32-call HCSE and 1/375 for 32-call label-free BBM. This is an empirical finite-suite certificate, not a worst-case approximation guarantee.

2 Preliminaries

Weighted graph and encoding tree. Let $G = (V, E, w)$ be an undirected graph with nonnegative edge weights. For $S \subseteq V$, let $V_S = \mathrm{vol}(S) = \sum_{u \in S} d_u$, and let g_S be the total weight of edges with exactly one endpoint in S. The total graph volume is $\mathrm{vol}(V) = \sum_u d_u = 2 \sum_{e \in E} w_e$. An encoding tree T has one leaf per vertex. Each internal node a represents the union S_a of its descendant leaves; a^- denotes its parent.

The structural entropy of G under T is

$$H^T(G) = -\sum_{a \neq r} \frac{g_{S_a}}{\mathrm{vol}(V)} \log_2 \frac{V_{S_a}}{V_{S_{a^-}}}. \tag{1}$$

Our sole optimization objective is Eq. (1); lower is better. No ground-truth label is used to construct or select a tree. For a zero-volume module, its summand is defined as zero; such a module has no incident positive-weight edge and can be attached after optimizing the positive-volume part of the tree.

Constructors. We distinguish a *constructor*, which produces an initial tree, from a *refiner*, which searches neighboring topologies. Our candidate pool contains (i) a binary SE-agglomerative dendrogram, (ii) a recursive SE partition tree, and (iii) a Paris dendrogram when the optional implementation is available [2]. HCSE and BBM are distinct external constructors and are evaluated from their official implementation [10].

Rooted NNI. An NNI move acts on a binary parent–child edge. Writing the local topology as $((A, B), C)$, the two nontrivial alternatives are $(A, (B, C))$ and $(B, (A, C))$. The leaf set, graph, and all modules outside the local parent $P = A \cup B \cup C$ remain unchanged. A rooted binary tree with n labeled leaves has finitely many topologies, namely $(2n - 3)!!$, so any strictly decreasing local search terminates.

3 NNI-Certified Entropy Search

Nest constructs several candidate hierarchies and refines each in a canonical binary neighborhood. Its local effect is exact, every committed endpoint is independently rescored, and small instances admit a separate global certificate.

3.1 A local structural-entropy identity

For disjoint sets X, Y, write $W(X, Y) = \sum_{u \in X, v \in Y} w_{uv}$. Consider the rotation shown in Fig. 1. Let $X = A \cup B$, $Y = B \cup C$, and $P = A \cup B \cup C$.

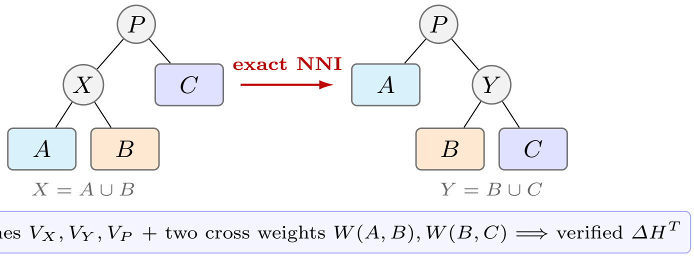

Fig. 1. A rooted NNI replaces only $X = A \cup B$ by $Y = B \cup C$. The highlighted local statistics determine the exact full-objective change; all leaves and the surrounding tree remain fixed.

Theorem 1 (Exact weighted NNI delta). *For an eligible rotation with $V_X, V_Y, V_P > 0$, the change in tree structural entropy is*

$$\Delta H^T = \frac{2}{\mathrm{vol}(V)} \left[W(A,B) \log_2 \frac{V_P}{V_X} + W(B,C) \log_2 \frac{V_Y}{V_P} \right]. \tag{2}$$

The symmetric alternative follows by exchanging A and B.

Proof. Deferred to the supplementary proof appendix.

Proposition 1 (Atomic advantage over merge–compress). *Let T_c contract the edge from P to $X = A \cup B$, exposing A, B, C as siblings, and let T' then merge B, C under $Y = B \cup C$. For these HCSE compress and merge primitives,*

$$H^{T_c}(G) - H^T(G) = \frac{2W(A,B)}{\mathrm{vol}(V)} \log_2 \frac{V_P}{V_X} \geq 0, \tag{3}$$

$$H^{T'}(G) - H^{T_c}(G) = -\frac{2W(B,C)}{\mathrm{vol}(V)} \log_2 \frac{V_P}{V_Y} \leq 0. \tag{4}$$

Their sum is Eq. (2)*. Thus NNI lowers entropy when the merge gain in Eq.* (4) *exceeds the compression loss in Eq.* (3)*, although neither desired reparenting nor an improving compression is available as one primitive.*

Proof. Deferred to the supplementary proof appendix.

This proves strict separation from monotone single-primitive search, not global dominance over a complete HCSE round, which may cross an intermediate increase. We evaluate Eq. (2) from cached sets and sparse adjacency lists, then verify every selected move by fully recomputing Eq. (1).

3.2 One-step certificate and compound escape

Algorithm 1: Exact NNI refinement of one candidate tree

Input: Graph G, tree T, tolerance ε, barrier β, beam b

repeat
- evaluate Eq. (2) for every legal NNI of T;
- apply the most negative move and verify the full objective;

until *no improving one-step move exists*;
for *at most* R *compound rounds* **do**
- retain the b first moves with smallest ΔH and $\Delta H \leq \beta$;
- expand every legal second NNI from each retained intermediate tree;
- **if** *no two-move endpoint has entropy below* $H^T(G)$ **then**
  - break;
- commit the best endpoint; rerun one-step descent;

return T

Proposition 2 (Monotonicity and local certificate). *One-step refinement never increases* $H^T(G)$. *If it stops before its move budget, the returned tree has no improving rooted NNI on any eligible binary parent–child edge.*

Proof. Deferred to the supplementary proof appendix.

Strict one-step descent can stop before a better tree separated by a shallow barrier. Algorithm 1 therefore keeps a bounded beam of first moves whose increase is at most β and evaluates a second NNI. It never commits the intermediate tree alone.

Proposition 3 (Safe compound search). *The compound stage never worsens its input. It can reach an improving endpoint even when every one-step neighbor is non-improving.*

Proof. Deferred to the supplementary proof appendix.

3.3 Fast NNI-certified entropy search

Nest applies Algorithm 1 independently to SE-agglomerative, recursive-SE, and Paris starts and returns the candidate with minimum verified entropy. The initializer is recorded for every output, so gains can be attributed to candidate selection, one-step refinement, and compound escape. With n leaves there are at most $2(n-2)$ oriented NNI alternatives per sweep. Our implementation uses best improvement, cached leaf sets, sparse cross-weight scans, and an $O(m \log h + n)$ full verification only for a selected move, where h is tree height. We make runtime claims empirically rather than assuming balanced trees.

Randomized basin coverage. Nest-R adds q coalescent starts, each built by repeatedly merging two uniformly selected current components, then applies the same NNI routine. The distribution is not uniform over labeled topologies. Selection uses only rescored entropy; increasing q cannot worsen the result, but gives no global guarantee.

3.4 Exact global audit on small graphs

Local certification does not imply global optimality. We therefore use a separate exact solver. SuperTAD proves binary sufficiency for general graphs, and HCSE also notes it [12, 10]; we restate the result to make the audit self-contained.

Proposition 4 (Binary sufficiency). *Every encoding tree has a binary refinement whose structural entropy is no greater. Consequently, a global minimum exists among rooted binary trees.*

Proof. Included for completeness in the supplementary proof appendix.

For a nonempty vertex subset S, let $F(S)$ be the minimum contribution of a binary subtree rooted at S, excluding the edge from S to its parent. Then

$$F(\{v\}) = 0,$$
$$F(S) = \min_{A \dot\cup B = S} \left\{ F(A) + F(B) - \frac{g_A}{\mathrm{vol}(V)} \log_2 \frac{V_A}{V_S} - \frac{g_B}{\mathrm{vol}(V)} \log_2 \frac{V_B}{V_S} \right\}. \tag{5}$$

Enumerating each unordered split once gives $O(3^n)$ time and $O(2^n)$ memory, an SE-specific instance of subset-trellis exact inference [8]. By Proposition 4, $F(V)$ is the global minimum over all encoding trees, not only over the binary subclass. This exponential routine is used only to audit 12-vertex instances; it is not part of Nest at deployment scale.

For an additional graph-only certificate, an edge-LCA expansion gives

$$H^T(G) = \sum_{uv \in E} \frac{w_{uv}}{\mathrm{vol}(V)} \log_2 \frac{V^2_{\mathrm{lca}(u,v)}}{d_u d_v} \geq \sum_{uv \in E} \frac{w_{uv}}{\mathrm{vol}(V)} \log_2 \frac{(d_u + d_v)^2}{d_u d_v}. \tag{6}$$

The inequality follows from $V_{\mathrm{lca}(u,v)} \geq d_u + d_v$. It is a universal lower bound, but the exact dynamic program gives the sharper certificates reported below.

4 Experiments

4.1 Questions and protocol

We ask: (Q1) how often do established constructors leave an improving NNI; (Q2) does compound search add improvements beyond a one-step certificate; (Q3) does lower entropy preserve or improve recovery of a planted hierarchy; and (Q4) how close is Nest to the global optimum on exactly solvable graphs?

The sealed main protocol uses 100 graph seeds per regime. Every method sees the identical weighted graph, and every tree is rescored by the same independent implementation of Eq. (1). We report paired means and 95% t-intervals, raw per-seed records, wall-clock time, and Python allocation peak. The compound search uses beam width 16, barrier $\beta = 0.05$ bits, and at most eight accepted compound rounds. No label is used by Nest for model selection. BBM is additionally shown with the planted number of fine clusters, which is an oracle advantage and is marked as such.

Data. Our controlled suite contains five two-level hierarchical stochastic block models: clean, noisy, imbalanced, weighted, and weak-hierarchy. Each balanced graph has 64 vertices, eight fine blocks, and four coarse blocks; the imbalanced regime has the same total size with block sizes from 4 to 12. The generator manifest, any 0.01-weight connectivity edges, and every seed are stored with the results. This family follows the hierarchical stochastic block model used to study recovery beyond worst-case inputs [3].

Methods. We compare SE agglomeration, a two-level Louvain tree [1], Paris [2], official HCSE, official BBM with oracle k [10], and NEST. We also refine each external constructor independently to measure its one-step and compound gap rather than crediting all gains to our preferred initializer.

Metrics. The primary metric is $H^T(G)$. Dendrogram purity is computed for the planted fine and coarse partitions: for every same-class vertex pair, it measures the class fraction under the pair's predicted LCA, then averages over pairs. We also report the fraction of runs improved by one-step and compound NNI, number of moves, runtime, and memory. For the exact suite we report the additive gap $H^T_{\mathrm{NEST}} - H^*$, relative gap $(H^T_{\mathrm{NEST}}/H^* - 1)100\%$, and *optimal-hit rate*: the fraction of graph instances whose objective is within $\max(10^{-9}, 10^{-8}|H^*|)$ of the exact optimum. This objective metric does not assert recovery of a unique planted tree.

4.2 A strict one-step trap

Before aggregate evaluation, an eight-vertex weighted witness verifies the purpose of compound search. Exact one-step descent stops at 1.970038 bits. Every one-NNI delta is nonnegative, but a first move crossing a 0.013970-bit barrier followed by a second move and ordinary descent reaches 1.920593 bits, a 0.049445-bit reduction. This witness is generated from a fixed seed and is covered by a regression test.

4.3 Frozen benchmark results

Table 1 reports the raw output of each method, before applying our audit to competing trees. NEST has the lowest mean H^T in all five regimes and on 500/500individual paired graphs. Its paired reduction is 0.5468 ± 0.0218 bits relative to HCSE and 0.5213 ± 0.0101 bits relative to BBM; it is lower on all 500 graphs against each method. Figure 2 shows paired evidence against the stronger of HCSE and BBM on each graph.

4.4 The operator audit

The constructor audit in Table 2 asks whether a returned tree can be improved by the exact local operator. One-step NNI improves SE agglomeration and oracle BBM on every graph. After one-step descent has certified its local neighborhood, bounded compound search still improves 62% of SE-agglomerative trees and 93%

Table 1. Tree structural entropy H^T over 100 graphs per regime (mean ± 95% CI; lower is better). BBM† receives the planted fine-cluster count.

Method	Clean	Noisy	Imbal.	Weighted	Weak
SE–agglomerative	2.899 ± 0.021	3.723 ± 0.017	3.162 ± 0.023	2.445 ± 0.016	3.461 ± 0.022
Louvain–2L	4.109 ± 0.022	4.603 ± 0.013	4.241 ± 0.019	3.657 ± 0.032	4.432 ± 0.018
Paris	2.869 ± 0.022	3.709 ± 0.017	3.129 ± 0.023	2.349 ± 0.015	3.438 ± 0.023
HCSE	3.228 ± 0.031	4.485 ± 0.068	3.598 ± 0.036	2.741 ± 0.029	4.032 ± 0.051
BBM†	3.335 ± 0.021	4.118 ± 0.020	3.573 ± 0.025	3.007 ± 0.020	3.924 ± 0.022
NEST (ours)	**2.827 ± 0.021**	**3.683 ± 0.017**	**3.086 ± 0.023**	**2.340 ± 0.015**	**3.414 ± 0.021**

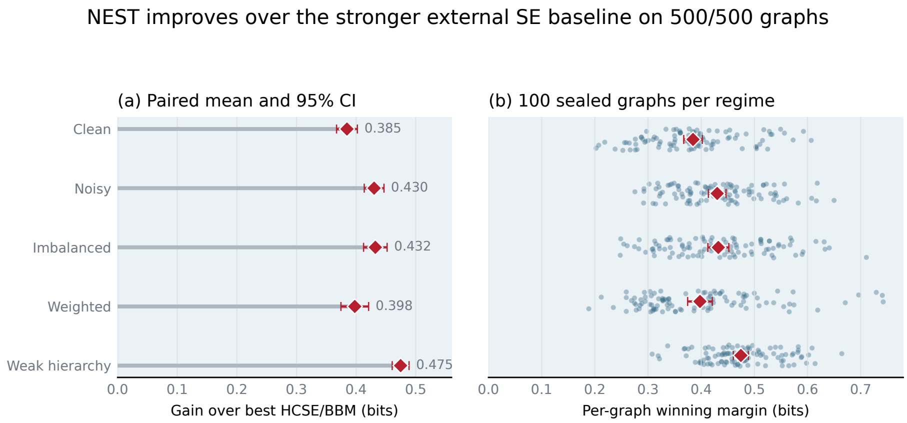

Fig. 2. Paired advantage over external structural-entropy methods. For each graph, the reference is the lower-entropy result of HCSE and BBM. (a) Regime-level mean reduction and 95% paired confidence interval. (b) Per-seed margins with small vertical jitter for visibility. NEST has lower entropy on all 500 graphs.

of BBM trees. Paris is closer to local optimality but is not immune: one-step and compound NNI improve 74% and 41% of its outputs. HCSE is usually locally stable, but still admits one-step and compound improvements on 3% and 4% of outputs. These observations support the neighborhood-separation claim without asserting that HCSE is generally weak.

4.5 Components and exact optimality

NEST forms a small candidate set from SE agglomeration, recursive SE, and Paris, refines each with the same exact operator, and returns the lowest independently rescored endpoint. The main benchmark contains 3,500 method records and 500 generator manifests. Its verifier checks hashes, schema, paired seeds, unique keys, and monotonicity of committed NNI endpoints.

On a separate frozen 50-graph subset, Table 3 separates the three algorithmic choices. The candidate pool improves 34 of 50 graphs over SE agglomeration;

Table 2. NNI audit pooled over 500 paired graphs. Reduction is relative to each constructor's raw H^T; rates are fractions of runs improved.

Constructor	1-NNI gain	1-NNI hit	2-step gain	2-step hit	Time
SE–agglomerative	1.97%	100%	0.14%	62%	0.642s
Paris	0.07%	74%	0.06%	41%	0.388s
HCSE	0.00%	3%	0.00%	4%	0.108s
BBM†	14.02%	100%	0.46%	93%	1.962s

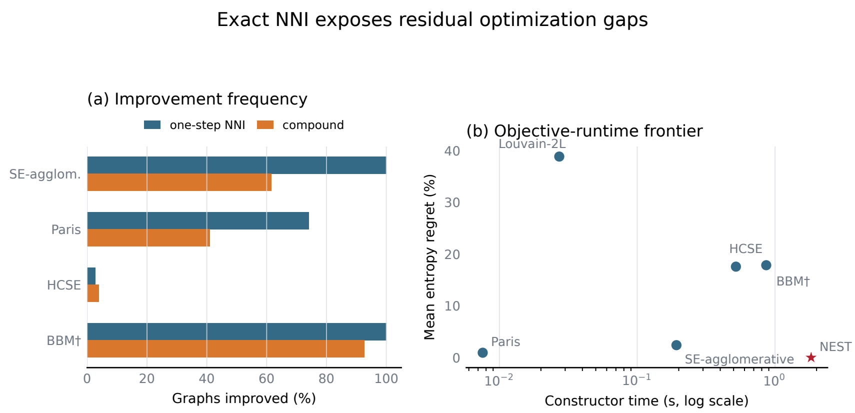

Fig. 3. Operator diagnostics pooled over 500 paired graphs. (a) Fraction of external constructor outputs improved by one-step NNI and by compound search after the one-step certificate. (b) Mean entropy regret versus constructor runtime; lower-left is better. Runtime includes Python allocation tracing and should be interpreted comparatively.

exact one-step NNI then improves 45 of 50 paired multi-start outputs; bounded compound search adds a smaller but nonzero gain on 30 graphs. Hence neither the initializer pool nor the escape stage alone explains the full result.

For the global audit, disjoint calibration fixes $q = 32$, after which we seal 250 graphs at $n = 12$, 100 at $n = 14$, and 25 at $n = 16$ across the same five regimes. Every restarted method receives 32 candidates. Selection uses only structural entropy; the exact dynamic program is consulted afterward. Table 4 shows that Nest reaches 370/375 optima, versus 2/375 for HCSE and 1/375 for label-free BBM. At $n = 16$, the 22/25 estimate has a wider exact binomial interval, which the table reports explicitly. All completed blocks have status zero, matching SHA-256 seals, complete seed coverage, finite outcomes, and independently recomputed summaries. This finite audit establishes neither a size-independent approximation ratio nor guaranteed global optimality beyond the declared instances.

Table 3. Component ablation pooled over 50 paired graphs. The gain and improved-run columns compare each row to the preceding row.

Variant	Mean H^T	Mean gain	Improved runs
SE agglomeration	3.1435	–	–
+ candidate pool	3.1041	0.0394	34/50
+ exact 1-NNI	3.0794	0.0247	45/50
+ compound escape	3.0759	0.0035	30/50

Table 4. Sealed exact-optimum audit with 32 candidates per method. Parentheses give exact 95% Clopper–Pearson intervals for the NEST hit rate. Gap is relative to H^*.

n	Graphs	NEST exact	95% CI	HCSE exact	BBM exact	Mean / max gap
12	250	**249/250**	[97.79, 99.99]%	2/250	1/250	0.00032% / 0.080%
14	100	**99/100**	[94.55, 99.97]%	0/100	0/100	0.00121% / 0.121%
16	25	**22/25**	[68.78, 97.45]%	0/25	0/25	0.03990% / 0.449%

4.6 Real networks

Table 5 complements the controlled suites with four networks bundled by NetworkX or the artifact library. NEST obtains the lowest raw entropy on 4/4networks. Because only Karate has a standard class label in the bundled loader and the remaining networks lack a unique ground-truth hierarchy, this experiment tests objective minimization rather than recovery.

5 Related Work

Structural entropy and hierarchy construction. Li and Pan introduced structural information as an encoding-tree measure of network organization [6]. HCSE and BBM develop hierarchy construction and balancing operations for this objective [10]. SuperTAD proves binary sufficiency and uses it in contiguity-constrained dynamic programs [12]; HCSE also notes the property [10]. Our exact audit instead optimizes arbitrary vertex subsets. The structural-entropy survey organizes the broader theory, optimization, and application literature [11]. NEST asks whether a returned tree is locally optimal under an exact topology rotation and uses the answer as both a diagnostic and an optimizer.

Graph hierarchical clustering. Paris is an agglomerative graph hierarchy related to modularity and provides a strong inexpensive initialization [2]. Louvain's multilevel aggregation optimizes modularity [1]; we evaluate its final partition as a two-level tree, not as an SE optimizer. Dasgupta-style objectives initiated a complementary algorithmic literature [4]; average linkage and divisive local search have guarantees for its revenue dual [9]. This objective is a useful neighbor, but its local delta cannot replace Eq. (1).

Table 5. Tree entropy on four bundled real networks (lower is better). BBM uses the ground-truth k when labels exist and Louvain's k otherwise.

Method	Karate	Florent.	Les Mis.	Davis
SE–agglomerative	2.698	1.923	3.078	3.055
Louvain–2L	3.405	2.452	3.923	3.813
Paris	2.597	1.923	2.920	2.981
HCSE	2.929	2.172	3.930	3.206
BBM	3.330	2.254	3.199	3.387
Nest (ours)	**2.586**	**1.919**	**2.916**	**2.962**

Exact inference and heuristic accuracy. Cluster trellises support exact hierarchical inference [8]. Our routine is an SE specialization: the prior binary-sufficiency property justifies the subset recurrence, while our arbitrary-subset formulation differs from SuperTAD's contiguity-constrained dynamic programs.

Local search in tree space. NNI is a classical neighborhood for phylogenetic trees; its reconfiguration distance is itself nontrivial [7]. Recent work studies NNI local search for hierarchical-clustering costs [5]. Our formula targets weighted graph structural entropy and exposes the two cross-module weights controlling a rotation. Unlike unit-clique landscape analyses, our method evaluates the complete objective on arbitrary weighted graphs. Its new content is the verified optimizer, compound search, multi-start construction, comparative evaluation, and exact global audit; local certification alone is not global optimality.

6 Limitations and Conclusion

NNI is a local operator, not a certificate of global optimality. Compound search explores only a bounded two-step neighborhood and does not exhaust all encoding trees. Multiway regions must also be resolved by a constructor before binary rotations can act on them. Lower structural entropy need not improve every external notion of hierarchy, which is why we report planted purity alongside the optimized objective. The exact solver is exponential and its near-optimality measurements therefore cover only the declared 12–16-vertex suites.

We introduced Nest, which derives an exact weighted NNI identity and uses it for monotone refinement, a local-optimality certificate, and safe bounded escape from shallow one-step traps. On the frozen five-regime suite, Nest achieved the lowest mean entropy in every regime and outperformed both HCSE and BBM on all 500 paired graphs. Prior binary sufficiency together with our exact arbitrary-subset dynamic program shows that 32-start Nest reaches the global optimum on 370/375 sealed instances across $n = 12, 14, 16$, versus 2/375 for HCSE and 1/375 for label-free BBM. This does not imply global optimality beyond the audited suites. The operator audit further shows that hierarchy construction, local optimality, and global optimality are distinct concerns. The result is a reproducible hierarchy-search framework whose guarantees and empirical gains can be checked directly.

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